

SWARNIM BARAPATRE

+91 8149 833 469 | Pune | swarnim335@gmail.com | github/swarnimcodes | linkedin/swarnimbarapatre

PROFESSIONAL SUMMARY

Full-Stack Software Engineer focused on backend-heavy product engineering, database-backed systems, and industrial software platforms. Experienced in building production APIs, data pipelines, graph-based workflows, CAD adapters, AI agents, and full-stack applications across TypeScript, Python, Node.js, and Go ecosystems. Strong bias toward maintainable architecture, performance-aware design, reliable data modeling, and practical automation.

EDUCATION

BITS Pilani
B.E. Chemical Engineering + M.Sc. Biological Sciences

Vasco da Gama, Goa
May, 2019 — July, 2024

PROFESSIONAL EXPERIENCE

Coditation Systems (Plant360.ai)
FullStack Software Engineer

Pune, Maharashtra
Feb, 2026 — Present

- Built an AI agent that generates equipment-isolation plans for real-world process scenarios in Plant360.
- Designed the agent around a self-orchestrating tool-calling loop, deterministic graph traversal modules, validation gates, evidence recovery through targeted follow-up calls, and auditable execution traces.
- Contributed to the Plant360.ai platform backend, developing production API services for industrial P&ID conversion and engineering-data workflows.
- Reverse-engineered SmartPlant™ P&ID automation APIs to build the SPPID-to-Plant360 adapter, extracting proprietary .pid drawings into Plant360's custom graph format through a REST API.
- Built dexpi-service as the next stage of the SmartPlant™ P&ID to DEXPI pipeline, converting Plant360 graph JSON into validated DEXPI output through async job processing, persisted job state, result retrieval APIs, bearer-token auth, and Docker deployment.
- Developed an AutoCAD adapter that renders Plant360 custom graph data into DWG drawings using reusable symbol catalogs, batch conversion scripts, and generated equipment, line, valve, instrument, tag, and annotation geometry.
- Built a custom symbol-conversion tool that automated a previously manual drafting workflow across 90+ symbol definitions, cutting 2 hours/day of repetitive effort down to a verification step.

AccurateIC
FullStack Software Engineer, AI Department

Pune, Maharashtra
Aug, 2024 — Feb, 2026

- Architected and developed ISACA, a full-stack monorepo for Indian Navy autonomous ship-control workflows, integrating ROS middleware through custom pub-sub transport and real-time multi-sensor fusion across radar, GPS, cameras, eMRU, and odometry streams.
- Built radar overlay, AIS tracking, dynamic route planning, waypoint management, voyage tracking, and shore-to-ship synchronization using bidirectional WebSocket architecture.
- Developed real-time shell command execution and monitoring flows with UI feedback, and set up Jenkins CI/CD with automated database migrations, seeding, and systemd orchestration.
- Led full-stack development for the BlueGulf x AccurateIC collaboration, adapting autonomous navigation systems for high-speed vessels and showcasing the platform at DriftX Abu Dhabi Autonomous Expo at Yas Marina.
- Architected and developed NeuroGen, an IoT-enabled genset monitoring platform with predictive maintenance, multi-variate anomaly detection, life expectancy simulations, and operational data visualizations.
- Drove architecture, code reviews, junior developer mentoring, and performance profiling, achieving 40% CPU reduction in critical application flows.

MasterSoft ERP Solutions
Software Engineering Intern, R&D Department

Nagpur, Maharashtra
Aug, 2023 — June, 2024

- Developed Panopticon, a multi-tenant employee productivity measurement system serving 700+ concurrent users, eliminating DeskTime dependency and saving Rs 53 Lakhs annually.
- Implemented Panopticon features including atomic app updates, live configuration sync, encrypted activity logs, and accurate productivity metrics for remote talent evaluation.
- Built an internal MS-SQL suite for database developers, reducing SQL task flows by 85% through stored procedure diffs, alter-script generation, and memory leak detection tooling.

- Built comprehensive internal developer toolchains and automation frameworks, reducing development cycle times by up to 14 days and improving team productivity across multiple projects.
- Maintained critical Linux server infrastructure, implemented proactive monitoring solutions, and resolved production incidents to ensure system uptime and reliability.

SKILLS

- **Languages:** TypeScript, JavaScript, Python, Go, C#, SQL, Bash, Powershell, Pascal
- **Frontend:** React, Zustand, Redux, Tanstack Query, HTML, CSS, TailwindCSS
- **Backend:** AdonisJS MVC, ExpressJS, NodeJS, OAuth, RabbitMQ, Lucid ORM, Flask, FastAPI, Django, Fiber, WebSockets, Server-Sent Events, Tool Calling,
- **Databases:** PostgreSQL, SQLite, Microsoft SQL Server, MongoDB
- **Cloud/DevOps:** AWS EC2, Docker, Nginx, GitLab, GitHub, GitHub Actions, Jenkins, Linux
- **Other:** ROS, Emacs, Vim, LaTeX, Typst

PERSONAL PROJECTS

c3fm — Terminal file manager in C3 Lang June, 2026 — present

- Terminal based file manager. Navigation using vim-like keys.
- Building from scratch without external libraries.
- Built using a low-level systems programming language with manual memory management using arenas.
- Tech Stack: *C3*: <https://c3-lang.org/>

Fero — Continuation of the RQuickShare Project May, 2026 — present

- Took up the continuation of an open-source project which was abandoned since 2 years used by thousands of people in the foss community.
- Implemented novel features leveraging the quickshare technology in-built in android devices for information transfer with minimal latency.
- Tech Stack: *Rust, Tauri, QuickShare, React, TypeScript, GitHub Actions*

Turepo — Package for Emacs Dec, 2025 — present

- Built a developer productivity package for the GNU Emacs text editor.
- Passed MELPA's code & quality standards. Package merged in MELPA.
- Tech Stack: *Emacs Lisp*

Kafkaesqueue — Custom Queueing System Oct, 2023 — June, 2024

- Lightweight queue mangement system. FIFO Guarantees.
- Opt-in parallel processing of tasks using threads. Cuts down processing time by $\mathcal{O}(\text{nproc})$.
- Built-in persistence via SQLite or in-memory storage.
- Tech Stack: *Python, FastAPI, Pydantic*.

EXTRA CURRICULARS

Swimming: National Swimmer. Multiple times State Gold Medalist.